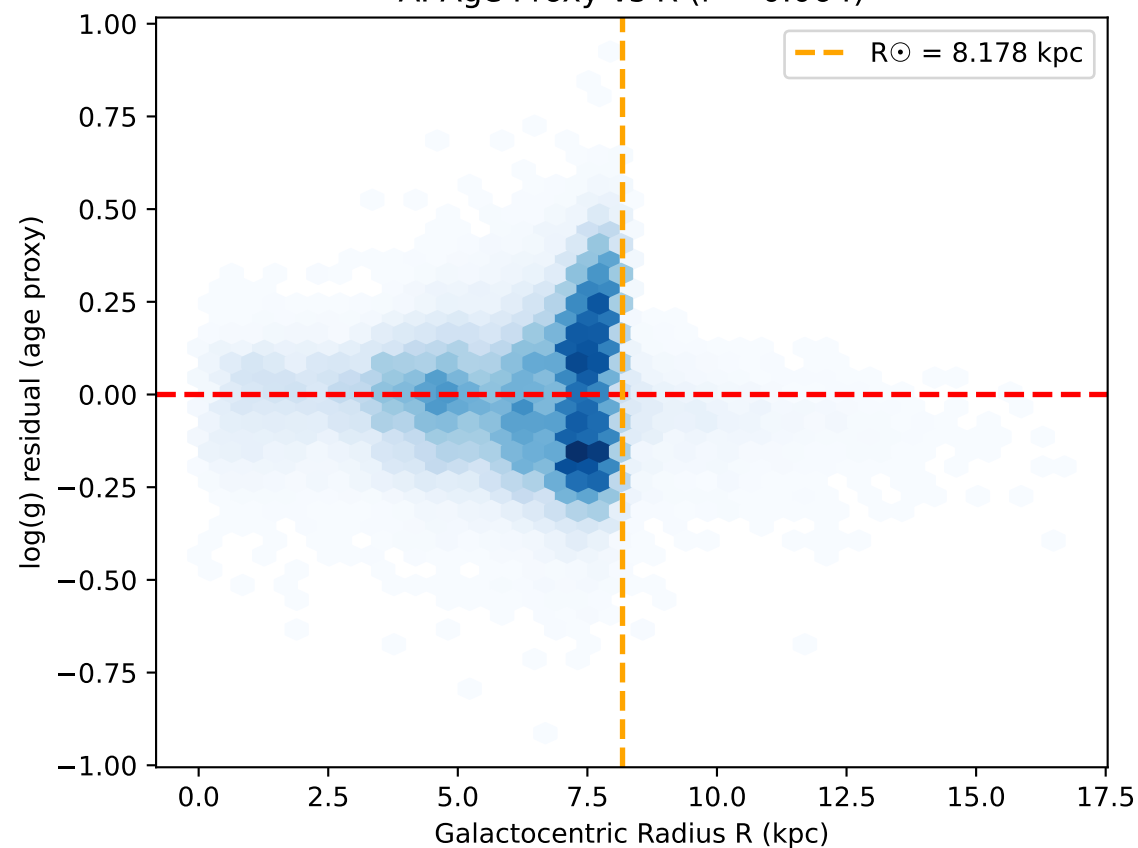
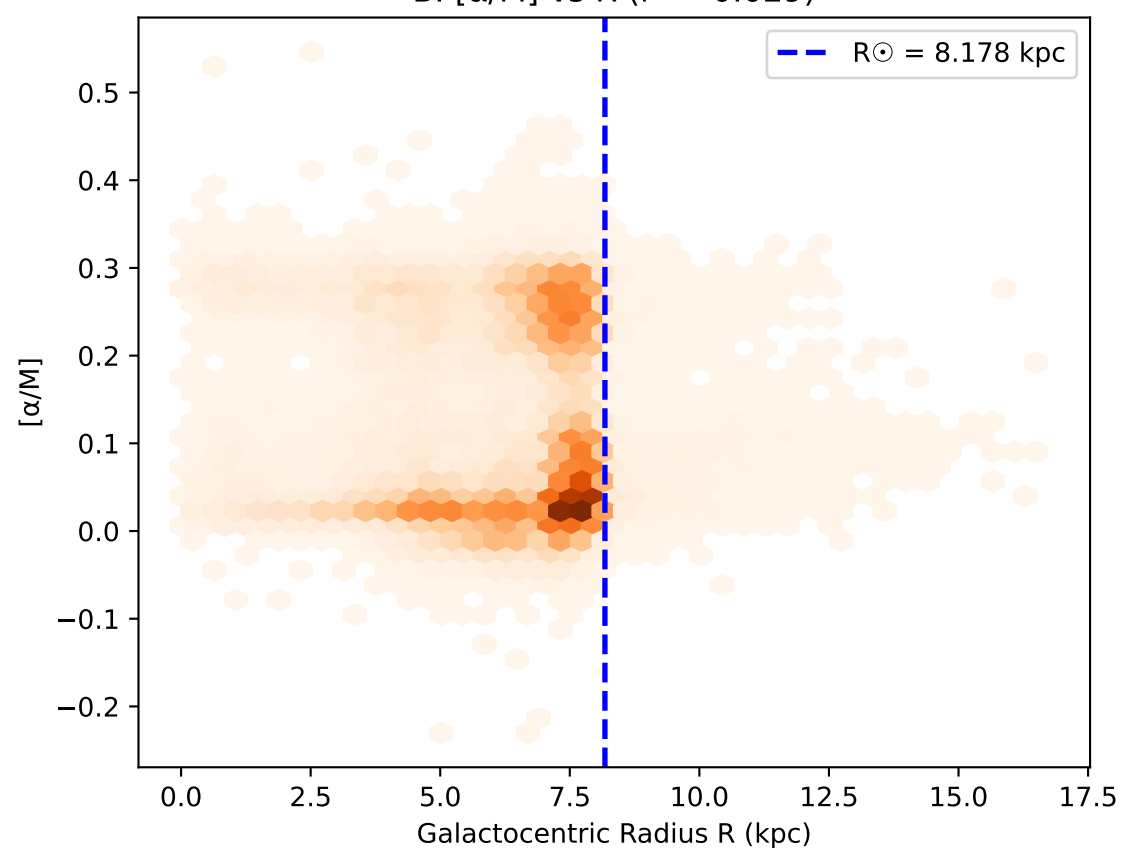
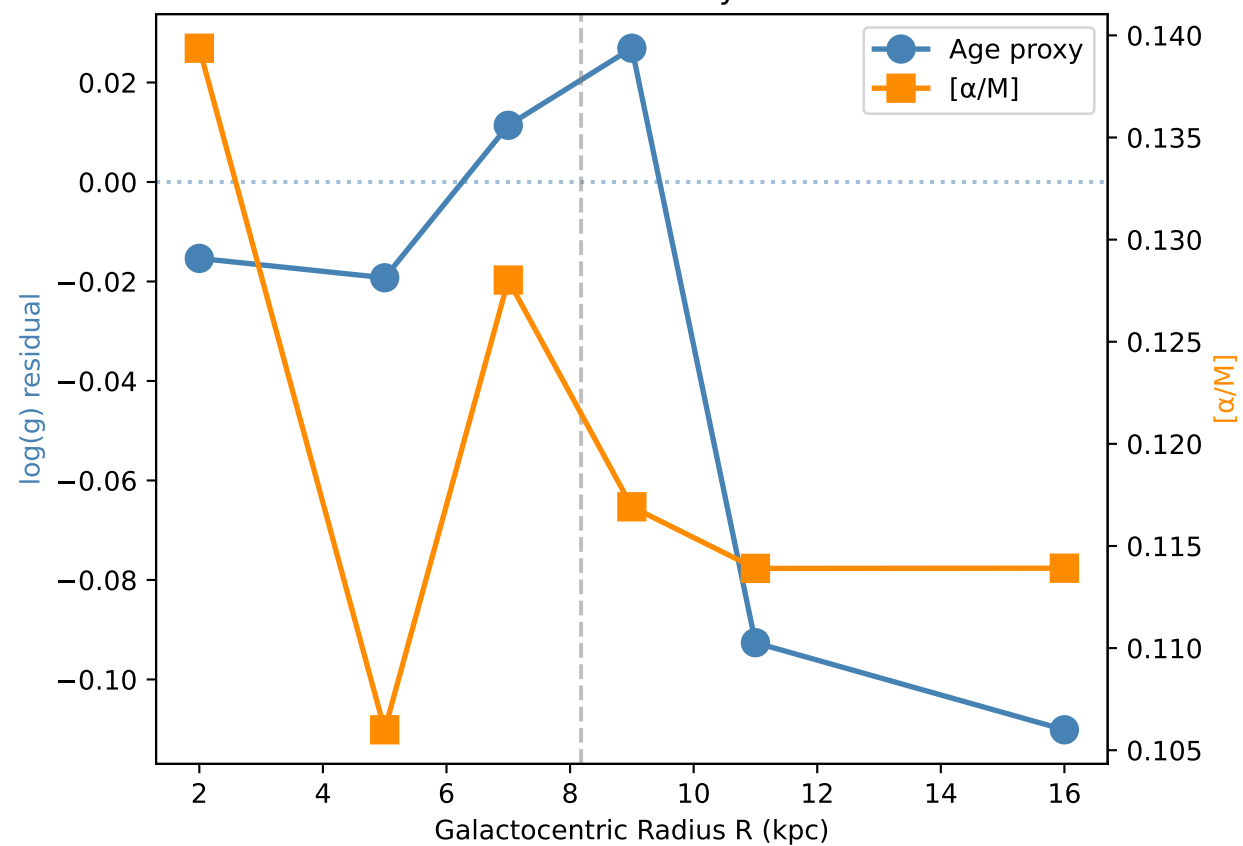


A. Age Proxy vs R ($r = 0.064$)B. $[\alpha/M]$ vs R ($r = -0.029$)

C. Binned Analysis



APOGEE STELLAR ARCHAEOLOGY TEST

Sample: 40,602 red giant stars
 R_{gc} range: 0.0 - 16.7 kpc

CORRELATIONS WITH R_{gc} :

Age proxy: $r = 0.0644$
 $[\alpha/M]$: $r = -0.0285$
 Discrepancy: $\Delta r = 0.0929$

PARTIAL CORRELATIONS (controlling for $[M/H]$):

Age proxy: $r = 0.0646$
 $[\alpha/M]$: $r = -0.1575$

CORRELATIONS WITH $\Delta\Phi/c^2$:

Age proxy: $r = -0.0579$
 $[\alpha/M]$: $r = 0.0476$

INTERPRETATION:

TEP_CONSISTENT

TEP PREDICTION:

Inner Galaxy stars (deeper Φ) should appear
 YOUNGER due to time dilation.